



Voltage and current debug signals rail to zero or max in mcb_pmsm_foc_qep_dyno_f28069m using LAUNCHXL-F28069M and BOOSTXL-DRV8305



Jonathan

📅 9 Mar 2026

💬 1 Answer

📅 Updated 12 Mar 2026

👁 41 Views (30 days)

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I am working with the Simulink example:

mcb_pmsm_foc_qep_dyno_f28069m

Hardware setup:

- Controller: LAUNCHXL-F28069M
- Inverters: 2 × BOOSTXL-DRV8305
- Two PMSMs coupled mechanically for dyno testing
- DC bus powered from an AC-DC wall adapter

The dyno control itself appears to work correctly. Motor 1 tracks speed commands and Motor 2 applies load torque as expected.

However I am having trouble with the electrical debug signals coming from the dyno model.

Problem

The debug signals for bus voltage and current appear incorrect:

- Values either collapse to zero or jump to the maximum value
- They do not track expected electrical behavior
- Mechanical signals such as speed, torque, and mechanical power appear normal and stable



Because the mechanical control loop works correctly, I suspect the issue may be somewhere in the ADC measurement path, scaling, or debug signal configuration rather than the control algorithm itself.

Question about the DC bus

Since the motors are coupled in a dyno configuration, Motor 2 can regenerate power back onto the DC bus. My supply is an AC-DC wall adapter which likely cannot sink current.

Could the lack of a regenerative path on the DC bus cause the voltage or current signals to saturate or behave abnormally?

Questions

1. Has anyone seen voltage or current debug signals rail to zero or full-scale in the mcb_pmsm_foc_qep_dyno_f28069m example?
2. Are there required ADC offset calibration or scaling steps when using BOOSTXL-DRV8305 boards?
3. Is it recommended to power the dyno setup from a battery or regenerative supply instead of a simple AC-DC adapter?

Any guidance on where to start debugging the measurement path would be greatly appreciated.

Thanks

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Answers (1)



Satyam on 12 Mar 2026

Edited: Satyam on 12 Mar 2026

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Hi Jonathan,

Since the dyno control and mechanical signals (speed/torque) behave correctly, the control algorithm and QEP feedback are likely fine, so the issue is most likely in the ADC measurement or scaling path.

- Check that the ADC channel mapping and scaling for phase currents and DC-bus voltage match the **BOOSTXL-**

DRV8305 hardware. Incorrect gain or channel configuration can cause debug signals to rail to 0 or full-scale. The expected hardware setup and scaling are described in the Motor Control Blockset example workflow: <https://www.mathworks.com/help/ti-c2000/ug/dual-motor-dyno-example.html>

- Ensure the ADC/current-sensor offset calibration runs correctly during initialization. If offsets are not calibrated or channels differ from the expected configuration in mcb_pmsm_foc_qep_dyno_f28069m, measured currents can appear saturated.
- In a dyno setup Motor-2 can regenerate power to the DC bus. If the AC-DC adapter cannot sink current, the DC-link voltage may rise or clamp, which can also lead to abnormal voltage/current readings. A battery or regenerative supply is typically recommended for dyno testing.

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Asked:

**Jonathan**

on 9 Mar 2026

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**Satyam**

on 12 Mar 2026

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